

ANSWERS

1.

- (a) (i) 0.8 cm B1
- (ii) (max.) kinetic energy = 2.56 mJ C1
 $v_{\text{(MAX)}} = \omega a$ C1
(max.) kinetic energy = $\frac{1}{2}m\omega^2 a^2$ or $\frac{1}{2}m\omega^2 (a^2 - x^2)$ C1
 $2.56 \times 10^{-3} = \frac{1}{2} \times 0.130 \times \omega^2 \times (0.8 \times 10^{-2})^2$ M1
 $\omega = 24.8 \text{ rad s}^{-1}$ C1
 $f = \omega/2\pi$ M1
= 4.0 Hz (3.95 Hz) A0
- (b) (i) line parallel to x-axis at 2.56 mJ B1
- (ii) 1 4.0 Hz B1
- 2 0.50 cm (allow $\pm 0.03 \text{ cm}$) B1

2.

- (a) (i) amplitude = 0.5 cm A1
- (ii) period = 0.8 s A1
- (b) (i) $\omega = 2\pi / T$ C1
= 7.85 rad s^{-1}
correct use of $v = \omega \sqrt{(x_0^2 - x^2)}$ B1
= $7.85 \times \sqrt{\{(0.5 \times 10^{-2})^2 - (0.2 \times 10^{-2})^2\}}$
= 3.6 cm s^{-1} A1
(if tangent drawn or clearly implied (B1)
 $3.6 \pm 0.3 \text{ cm s}^{-1}$ (A2)
but allow 1 mark for $> \pm 0.3$ but $\leq \pm 0.6 \text{ cm s}^{-1}$)
- (ii) $d = 15.8 \text{ cm}$ A1
- (c) (i) (continuous) loss of energy / reduction in
amplitude (from the oscillating system) B1
caused by force acting in opposite direction to the motion / friction /
viscous forces B1
- (ii) same period / small increase in period B1
line displacement always less than that on Fig.3.2 (ignore first $T/4$) M1
peak progressively smaller A1

3.

- (a) acceleration / force (directly) proportional to displacement M1
 and *either* directed towards fixed point
 or acceleration & displacement in opposite directions A1

- (b) (i) maximum / minimum height / 8 mm above cloth / 14 mm below cloth B1

(ii) 1. $a = 11 \text{ mm}$ A1

2. $\omega = 2\pi f$ C1

$= 2\pi \times 4.5$

$= 28.3 \text{ rad s}^{-1}$ (do not allow 1 s.f.) A1

(c) (i) $v = \omega a$ C1

$= 28.3 \times 11 \times 10^{-3}$

$= 0.31 \text{ m s}^{-1}$ (do not allow 1 s.f.) A1

(ii) $v = \omega \sqrt{a^2 - y^2}$

$y = 3 \text{ mm}$ C1

$= 28.3 \times 10^{-3} \sqrt{11^2 - 3^2}$ C1

$= 0.30 \text{ m s}^{-1}$ (allow 1 s.f.) A1

4.

(a) (i) $(\theta =) \omega t$ (allow any subject if all terms given) B1

(ii) $(SQ =) r \sin \omega t$ (allow any subject if all terms given) B1

(b) this is the solution of the equation $a = -\omega^2 x$ M1

$a = -\omega^2 x$ is the (defining) equation of s.h.m. A1

(c) (i) $f = \omega / 2\pi$ C1

$= 4.7 / 2\pi$

$= 0.75 \text{ Hz}$ A1

(ii) $v = r\omega$ (r must be identified) C1

$= 4.7 \times 12$

$= 56 \text{ cm s}^{-1}$ A1

5.

- (a) (i) amplitude = 0.2 mm A1
- (ii) period = 1.2 ms C1
frequency = 830 Hz A1
- (b) (i) any two of zero, 0.6 ms and 1.2 ms A1
- (ii) any two of 0.3 ms, 0.9 ms, 1.5 ms A1
- (c) either $v = \omega x_0 = 2\pi f x_0$
 $= 2\pi \times 830 \times 0.2 \times 10^{-3} = 1.05 \text{ m s}^{-1}$
 or slope of graph = 1.0 m s^{-1} (allow $\pm 0.1 \text{ m s}^{-1}$) C1
 $E_K = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 2.5 \times 10^{-3} \times 1.05^2$ C1
 $= 1.4 \times 10^{-3} \text{ J}$ A1
- (d) (i) large / maximum amplitude of vibration B1
 when impressed frequency equals natural frequency of vibration B1
- (ii) e.g. metal panels on machinery vibrate / oscillate (M1)
 motor in machine impresses frequency on panel (A1)
 e.g. car suspension system vibrates / oscillates (M1)
 going over bumps would give large amplitude vibrations (A1)
 any feasible example, M1 + A1

6.

- (a) straight line through origin B1
 negative gradient B1
- (b) $a = -\omega^2 x$ and $\omega = 2\pi f$ C1
 $750 = (2\pi f)^2 \times 0.3 \times 10^{-3}$ C1
 $f = 250 \text{ Hz}$ A1
- (c) straight line between (-0.3, +190) and (+0.3, -190) A2
 (allow 1 mark for end of line incorrect by one grid square or line does not extend to $\pm 0.3 \text{ mm}$)

7.

- (a) (i) to-and-fro / backward and forward motion (between two limits) B1
- (ii) no energy loss or gain / no external force acting / constant energy B1
- (iii) acceleration directed towards a fixed point B1
acceleration proportional to distance from the fixed point / displacement B1
- (b) acceleration is constant (magnitude) M1
so cannot be s.h.m. A1

8.

- (a) (i) reduction in energy (of the oscillations) (B1)
reduction in amplitude / energy of oscillations (B1)
due to force (always) opposing motion / resistive forces (B1)
any two of the above, max 2
- (ii) amplitude is decreasing (very) gradually / oscillations would continue (for a long time) / many oscillations M1
light damping A1
- (b) (i) frequency = $1 / 0.3$
= 3.3 Hz A1
allow points taken from time axis giving $f = 3.45$ Hz
- (ii) energy = $\frac{1}{2}mv^2$ and $v = \omega a$ C1
= $\frac{1}{2} \times 0.065 \times (2\pi/0.3)^2 \times (1.5 \times 10^{-2})^2$ M1
= 3.2 mJ A0
- (c) amplitude reduces exponentially / does not decrease linearly M1
so will be not be 0.7 cm A1

9.

- (a) (i) resonance B1
- (ii) amplitude 16 mm and frequency 4.6 Hz A1
- (b) (i) $a = (-)\omega^2 x$ and $\omega = 2\pi f$ C1
 $a = 4\pi^2 \times 4.6^2 \times 16 \times 10^{-3}$ C1
 $= 13.4 \text{ ms}^{-2}$ A1
- (ii) $F = ma$ C1
 $= 150 \times 10^{-3} \times 13.4$
 $= 2.0 \text{ N}$ A1
- (c) line always 'below' given line and never zero M1
peak is at 4.6 Hz (or slightly less) and flatter A1

10.

- (a) (i) 8.0 cm A1
- (ii) $2\pi f = 220$ C1
 $f = 35$ (condone unit) A1
- (iii) line drawn mid-way between AB and CD (allow $\pm 2 \text{ mm}$) B1
- (iv) $v = \omega a$ C1
 $= 220 \times 4.0$
 $= 880 \text{ cm s}^{-1}$ A1
- (b) (i) 1. line drawn 3 cm above AB (allow $\pm 2 \text{ mm}$) B1
2. arrow pointing upwards B1
- (ii) 1. line drawn 3 cm above AB (allow $\pm 2 \text{ mm}$) B1
2. arrow pointing downwards B1
- (iii) $v = \omega \sqrt{a^2 - x^2}$
 $= 220 \times \sqrt{4.0^2 - 2.0^2}$ C1
 $= 760 \text{ cm s}^{-1}$ A1
(incorrect value for x, 0/2 marks)

11.

- (a) acceleration / force proportional to displacement from a fixed point M1
 acceleration / force (always) directed towards that fixed point / in opposite direction to displacement A1
- (b) (i) $A\rho g / m$ is a constant and so acceleration proportional to x B1
 negative sign shows acceleration towards a fixed point / in opposite direction to displacement B1
- (ii) $\omega^2 = (A\rho g / m)$ C1
 $\omega = 2\pi f$ C1
 $(2 \times \pi \times 1.5)^2 = (\{4.5 \times 10^{-4} \times 1.0 \times 10^3 \times 9.81\} / m)$ C1
 $m = 50 \text{ g}$ A1

12.

- (a) (i) amplitude remains constant B1
- (ii) amplitude decreases gradually M1
 light damping A1
- (iii) period = 0.80 s C1
 frequency = 1.25 Hz (period not 0.8 s, then 0/2) A1
- (b) (i) (induced) e.m.f. is proportional to M1
 rate of change/cutting of (magnetic) flux (linkage) A1
- (ii) a current is induced in the coil M1
 as magnet moves in coil A1
 current in resistor gives rise to a heating effect M1
 thermal energy is derived from energy of oscillation of the magnet A1

13.

- (a) acceleration proportional to displacement/distance from fixed point M1
 and in opposite directions/directed towards fixed point A1
- (b) energy = $\frac{1}{2}m\omega^2x_0^2$ and $\omega = 2\pi f$ C1
 $= \frac{1}{2} \times 5.8 \times 10^{-3} \times (2\pi \times 4.5)^2 \times (3.0 \times 10^{-3})^2$ C1
 $= 2.1 \times 10^{-5} \text{ J}$ A1
- (c) (i) at maximum displacement M1
 above rest position A1
- (ii) acceleration = $(-)\omega^2x_0$ and acceleration = 9.81 or g C1
 $9.81 = (2\pi \times 4.5)^2 \times x_0$
 $x_0 = 1.2 \times 10^{-2} \text{ m}$ A1

14.

(a) $a = (-)\omega^2 x$ and $\omega = 2\pi/T$ C1
 $T = 0.60 \text{ s}$ C1
 $a = (4\pi^2 \times 2.0 \times 10^{-2}) / (0.6)^2$
 $= 2.2 \text{ ms}^{-2}$ A1

(b) sinusoidal wave with all values positive B1
 all values positive, all peaks at E_K and energy = 0 at $t = 0$ B1
 period = 0.30 s B1

15.

(a) energy = $\frac{1}{2}m\omega^2 a^2$ and $\omega = 2\pi f$ C1
 $= \frac{1}{2} \times 37 \times 10^{-3} \times (2\pi \times 3.5)^2 \times (2.8 \times 10^{-2})^2$ M1
 $= 7.0 \times 10^{-3} \text{ J}$ A0
 (allow $2\pi \times 3.5$ shown as 7π)

Energy = $\frac{1}{2}mv^2$ and $v = r\omega$ (C1)
 Correct substitution (M1)
 Energy = $7.0 \times 10^{-3} \text{ J}$ (A0)

(b) $E_K = E_P$
 $\frac{1}{2}m\omega^2(a^2 - x^2) = \frac{1}{2}m\omega^2 x^2$ or E_K or $E_P = 3.5 \text{ mJ}$ C1
 $x = a/\sqrt{2} = 2.8/\sqrt{2}$ or $E_K = \frac{1}{2}m\omega^2(a^2 - x^2)$ or $E_P = \frac{1}{2}m\omega^2 x^2$ C1
 $= 2.0 \text{ cm}$ A1
 (E_K or $E_P = 7.0 \text{ mJ}$ scores 0/3)

Allow: $k = 17.9$ (C1)
 $E = \frac{1}{2}kx^2$ (C1)
 $x = 2.0 \text{ cm}$ (A1)

(c) (i) graph: horizontal line, y-intercept = 7.0 mJ with end-points of line at +2.8 cm and -2.8 cm B1

(ii) graph: reasonable curve B1
 with maximum at (0, 7.0) end-points of line at (-2.8, 0) and (+2.8, 0) B1

(iii) graph: inverted version of (ii) M1
 with intersections at (-2.0, 3.5) and (+2.0, 3.5) A1
 (Allow marks in (iii), but not in (ii), if graphs K & P are not labelled)

(d) gravitational potential energy B1

16.

- (a) straight line through origin M1
shows acceleration proportional to displacement A1
negative gradient M1
shows acceleration and displacement in opposite directions A1
- (b) (i) 2.8 cm A1
- (ii) *either* gradient = ω^2 and $\omega = 2\pi f$ or $a = -\omega^2 x$ and $\omega = 2\pi f$ C1
gradient = $13.5 / (2.8 \times 10^{-2}) = 482$
 $\omega = 22 \text{ rad s}^{-1}$ C1
frequency = $(22/2\pi =) 3.5 \text{ Hz}$ A1
- (c) e.g. lower spring may not be extended
e.g. upper spring may exceed limit of proportionality / elastic limit
(*any sensible suggestion*) B1

17.

- (a) (i) 1. 0.1 s, 0.3 s, 0.5 s, etc (*any two*) A1
2. either 0, 0.4 s, 0.8 s, 1.2 s
or
0.2 s, 0.6 s, 1.0 s (*any two*) A1
- (ii) period = 0.4 s C1
frequency = $(1/0.4 =) 2.5 \text{ Hz}$ A1
- (iii) phase difference = 90° or $\frac{1}{2} \pi \text{ rad}$ B1
- (b) frequency = 2.4 – 2.5 Hz B1
- (c) e.g. attach sheet of card to trolley M1
increases damping / frictional force A1
e.g. reduce oscillator amplitude (M1)
reduces power/energy input to system (A1)

18.

- (a) (i) 1. amplitude = 1.7 cm A1
2. period = 0.36 cm C1
frequency = $1/0.36$
frequency = 2.8 Hz A1
- (ii) $a = (-)\omega^2 x$ and $\omega = 2\pi/T$ C1
acceleration = $(2\pi/0.36)^2 \times 1.7 \times 10^{-2}$ M1
= 5.2 m s^{-2} A0
- (b) graph: straight line, through origin, with negative gradient M1
from $(-1.7 \times 10^{-2}, 5.2)$ to $(1.7 \times 10^{-2}, -5.2)$ A1
(if scale not reasonable, do not allow second mark)
- (c) either kinetic energy = $\frac{1}{2}m\omega^2(x_0^2 - x^2)$
or potential energy = $\frac{1}{2}m\omega^2 x^2$ and potential energy = kinetic energy B1
 $\frac{1}{2}m\omega^2(x_0^2 - x^2) = \frac{1}{2} \times \frac{1}{2}m\omega^2 x_0^2$ or $\frac{1}{2}m\omega^2 x^2 = \frac{1}{2} \times \frac{1}{2}m\omega^2 x_0^2$ C1
 $x_0^2 = 2x^2$
 $x = x_0 / \sqrt{2} = 1.7 / \sqrt{2}$
= 1.2 cm A1

19.

- (a) (i) $\omega = 2\pi / T$
= $2\pi / 0.69$ C1
= 9.1 rad s^{-1} A1
(allow use of $f = 1.5 \text{ Hz}$ to give $\omega = 9.4 \text{ rad s}^{-1}$)
- (ii) 1. $x = 2.1 \cos 9.1t$
2.1 and 9.1 numerical values B1
use of cos B1
2. $v_0 = 2.1 \times 10^{-2} \times 9.1$ (allow ecf of value of x_0 from (ii)1.)
= 0.19 m s^{-1} B1
 $v = v_0 \sin 9.1t$ (allow $\cos 9.1t$ if $\sin$ used in (ii)1.) B1
- (b) energy = either $\frac{1}{2}mv_0^2$ or $\frac{1}{2}m\omega^2 x_0^2$
= either $\frac{1}{2} \times 0.078 \times 0.19^2$ or $\frac{1}{2} \times 0.078 \times 9.1^2 \times (2.1 \times 10^{-2})^2$ C1
= $1.4 \times 10^{-3} \text{ J}$ A1

20.

- (a) (i) any two from 0.3(0) s, 0.9(0) s, 1.50 s (*allow 2.1 s etc.*) B1
- (ii) *either* $v = \omega x$ and $\omega = 2\pi/T$ C1
 $v = (2\pi/1.2) \times 1.5 \times 10^{-2}$ M1
 $= 0.079 \text{ m s}^{-1}$ A0
or gradient drawn clearly at a correct position (C1)
 working clear (M1)
 to give $(0.08 \pm 0.01) \text{ m s}^{-1}$ (A0)
- (b) (i) sketch: curve from $(\pm 1.5, 0)$ passing through $(0, 25)$ M1
 reasonable shape (*curved with both intersections between*
 $y = 12.0 \rightarrow 13.0$) A1
- (ii) at max. amplitude potential energy is total energy B1
 total energy = 4.0 mJ B1

21.

- (a) kinetic (energy)/KE/ E_k B1
- (b) *either* change in energy = 0.60 mJ
or max E proportional to (amplitude)²/equivalent numerical working B1
 new amplitude is 1.3 cm B1
 change in amplitude = 0.2 cm B1

22.

- (a) acceleration/force proportional to displacement (from a fixed point) M1
either acceleration and displacement in opposite directions
or acceleration always directed towards a fixed point A1
- (b) (i) g and r are constant so a is proportional to x B1
 negative sign shows a and x are in opposite directions B1
- (ii) $\omega^2 = g/r$ and $\omega = 2\pi/T$ C1
 $\omega^2 = 9.8/0.28$
 $= 35$ C1
 $T = 2\pi/\sqrt{35} = 1.06 \text{ s}$
 time interval $\tau = 0.53 \text{ s}$ A1
- (c) sketch: time period constant (or increases very slightly) M1
 drawn line always 'inside' given loops A1
 successive decrease in peak height A1

23.

- (a) acceleration/force proportional to displacement (from a fixed point) M1
either acceleration and displacement in opposite directions A1
or acceleration always directed towards a fixed point

- (b) (i) zero & 0.625 s *or* 0.625 s & 1.25 s *or* 1.25 s & 1.875 s *or* 1.875 s & 2.5 s A1

- (ii) 1. $\omega = 2\pi / T$ *and* $v_0 = \omega x_0$ C1
 $\omega = 2\pi / 1.25$
 $= 5.03 \text{ rad s}^{-1}$ C1

$$v_0 = 5.03 \times 3.2$$

$$= 16.1 \text{ cm s}^{-1} \text{ (allow 2 s.f.)} \quad \text{A1}$$

2. $v = \omega \sqrt{(x_0^2 - x^2)}$
either $\frac{1}{2}\omega a = \omega \sqrt{(x_0^2 - x^2)}$ *or* $\frac{1}{2} \times 16.1 = 5.03 \sqrt{(3.2^2 - x^2)}$ C1
 $x_0^2 / 4 = x_0^2 - x^2$ $2.58 = 3.2^2 - x^2$
 $x = 2.8 \text{ cm}$ $x = 2.8 \text{ cm}$ A1

- (c) sketch: loop with origin at its centre M1
correct intercepts & shape based on (b)(ii) A1

24.

- (a) (i) frequency at which object is made to vibrate/oscillate B1
(ii) frequency at which object vibrates when free to do so B1
(iii) maximum amplitude of vibration of oscillating body B1
when forced frequency equals natural frequency (of vibration) B1

- (b) e.g. vibration of quartz/piezoelectric crystal (*what is vibrating*) M1
either for accurate timing
or maximise amplitude of ultrasound waves (*why it is useful*) A1

- (c) e.g. vibrating metal panels (*what is vibrating*) M1
either place strengthening struts across the panel
or change shape/area of panel (*how it is reduced*) A1

25.

- (a) displacement (directly) proportional to acceleration/force M1
either displacement and acceleration in opposite directions
or acceleration (always) towards a (fixed) point A1

- (b) (i) $\frac{1}{3}\pi$ rad or 1.05 rad (*allow 60° if unit clear*) A1
- (ii) $a_0 = -\omega^2 x_0$
 $= (-) (2\pi/1.2)^2 \times 0.030$ C1
 $= (-) 0.82 \text{ m s}^{-2}$ A1
(special case: using oscillator P gives $x_0 = 1.7 \text{ cm}$ and $a_0 = 0.47 \text{ m s}^{-1}$ for 1/2)
- (iii) max. energy $\propto x_0^2$
ratio = $3.0^2/1.7^2$ C1
 $= 3.1$ (*at least 2 s.f.*) A1
(if has inverse ratio but has stated max. energy $\propto x_0^2$ then allow 1/2)
- (c) graph: straight line through (0,0) with negative gradient M1
correct end-points $(-3.0, +0.82)$ and $(+3.0, -0.82)$ A1
- 26.
- (a) free: (body oscillates) without any loss of energy/no resistive forces/no external forces applied B1
forced: continuous energy input (required)/body is made to vibrate by an (external) periodic force/driving oscillator B1
- (b) (i) idea of resonance B1
maximum amplitude at natural frequency B1
frequency = 2.1 Hz (*allow 2.08 to 2.12 Hz*) B1
- (ii) peak not very sharp/amplitude not infinite so frictional forces are present B1
- (c) $v = \omega x_0$
 $= 2\pi \times 2.1 \times 4.7 \times 10^{-2}$ (*allow ECF from (b)(i)*) C1
 $= 0.62 \text{ m s}^{-1}$ A1
- 27.
- (a) acceleration/force proportional to distance from a fixed point or displacement M1
either acceleration/force and displacement in opposite directions
or acceleration/force (always) directed towards a fixed point/mean position/equilibrium position A1
- (b) $h\rho g = Mg/A$ B1
 $h \times 790 \times 4.9 \times 10^{-4} = 70 \times 10^{-3}$ leading to $h = 0.18 \text{ m}$ or 18 cm A1

(c) (i) 1. $\omega^2 = (790 \times 4.9 \times 10^{-4} \times 9.81) / (70 \times 10^{-3})$ C1
 $= 54.25$

$\omega = 7.37 \text{ (rad s}^{-1}\text{)}$
 period $(= 2\pi / \omega) = 0.85 \text{ s}$ C1

$t_1 = 0.43 \text{ s}$ A1

2. $t_3 = 1.28 \text{ s (allow 2 s.f.)}$ A1

(ii) energy of peak $= \frac{1}{2} M \omega^2 x_0^2$ B1

change $= \frac{1}{2} \times 70 \times 10^{-3} \times 54.25 \{ (2.2 \times 10^{-2})^2 - (1.0 \times 10^{-2})^2 \}$ C1
 $= 7.3 \times 10^{-4} \text{ J}$ A1

28.

(a) amplitude $= 1.8 \text{ cm}$ and period $= 0.30 \text{ s}$ A1

(b) $E_K = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$ or $E_K = \frac{1}{2} m v^2$ and $v = \pm \omega \sqrt{(x_0^2 - x^2)}$ C1
 $= \frac{1}{2} \times 0.080 \times (2\pi / 0.30)^2 \times [(1.8 \times 10^{-2})^2 - (1.2 \times 10^{-2})^2]$ C1
 $= 3.2 \times 10^{-3} \text{ J}$ A1

29.

(a) acceleration/force proportional to displacement (from fixed point) M1

acceleration/force and displacement in opposite directions A1

(b) maximum displacements/accelerations are different B1

graph is curved/not a straight line B1

(c) (i) $\omega = 2\pi / T$ and $T = 0.8 \text{ s}$ C1

$\omega = 7.9 \text{ rad s}^{-1}$ A1

(ii) $a = (-)\omega^2 x$
 $= 7.85^2 \times 1.5 \times 10^{-2}$ C1

$= 0.93 \text{ ms}^{-2}$ or 0.94 ms^{-2} A1

(iii) $\Delta E = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$ C1

$= \frac{1}{2} \times 120 \times 10^{-3} \times 7.85^2 \times \{ (1.5 \times 10^{-2})^2 - (0.9 \times 10^{-2})^2 \}$ C1

$= 5.3 \times 10^{-4} \text{ J}$ A1

30.

(a) (i) $T = 0.60 \text{ s}$ and $\omega = 2\pi / T$ C1

$$\omega = 10(10.47) \text{ rad s}^{-1} \quad \text{A1}$$

(ii) energy = $\frac{1}{2}m\omega^2 x_0^2$ or $\frac{1}{2}mv^2$ and $v = \omega x_0$ C1

$$= \frac{1}{2} \times 120 \times 10^{-3} \times (10.5)^2 \times (2.0 \times 10^{-2})^2$$

$$= 2.6 \times 10^{-3} \text{ J} \quad \text{A1}$$

(b) sketch: smooth curve in correct directions B1

peak at f M1

amplitude never zero and line extends from $0.7f$ to $1.3f$ A1

(c) sketch: peaked line always below a peaked line A M1

peak not as sharp and at (or slightly less than) frequency of peak in line A A1

31.

(a) $2k/m = \omega^2$ M1

$$\omega = 2\pi f \quad \text{M1}$$

$$(2 \times 64 / 0.810) = (2\pi \times f)^2 \text{ leading to } f = 2.0 \text{ Hz} \quad \text{A1}$$

(b) $v_0 = \omega x_0$ or $v_0 = 2\pi f x_0$

or

$$v = \omega(x_0^2 - x^2)^{1/2} \text{ and } x = 0 \quad \text{C1}$$

$$v_0 = 2\pi \times 2.0 \times 1.6 \times 10^{-2}$$

$$= 0.20 \text{ m s}^{-1} \quad \text{A1}$$

(c) frequency: reduced/decreased B1

maximum speed: reduced/decreased B1

32.

(a) (i) 0.225 s and 0.525 s A1

(ii) period or $T = 0.30$ s and $\omega = 2\pi / T$ C1

$$\omega = 2\pi / 0.30$$

$$\omega = 21 \text{ rad s}^{-1} \quad \text{A1}$$

(iii) speed = ωx_0 or $\omega(x_0^2 - x^2)^{1/2}$ and $x = 0$ C1

$$= 20.9 \times 2.0 \times 10^{-2} = 0.42 \text{ ms}^{-1} \quad \text{A1}$$

or

use of tangent method:

correct tangent shown on Fig. 4.2

(C1)

working e.g. $\Delta y / \Delta x$ leading to maximum speed in range (0.38–0.46) ms^{-1}

(A1)

(b) sketch: reasonably shaped continuous oval/circle surrounding (0,0) B1

curve passes through (0, 0.42) and (0, –0.42) B1

curve passes through (2.0, 0) and (–2.0, 0) B1

33.

3(a)	m is constant or k/m is constant <u>and</u> so acceleration / a proportional to displacement / x	B1
	negative sign shows that acceleration / a is in opposite direction to displacement / x	B1
	or negative sign shows acceleration / a is towards fixed point	
3(b)	evidence of comparison to expression to $a = -\omega^2 x$	B1
	$\omega^2 = k/m$ or $\omega^2 = 4.0/m$ hence $\omega = 2.0/\sqrt{m}$	A1
3(c)	$E_K = \frac{1}{2} m \omega^2 x_0^2$ or $E_K = \frac{1}{2} m v^2$ <u>and</u> $v = \omega x_0$	C1
	$= \frac{1}{2} m (4.0/m) (3.0 \times 10^{-2})^2$	C1
	$= 1.8 \times 10^{-3} \text{ J}$	A1
3(d)	new $x_0 = \sqrt{[(1.8 \times 10^{-3} / 2) \times (2 / m \times (m / 4.0))]}$	C1
	or ($E_K \propto x_0^2$ so) new $x_0 = \sqrt{[\frac{1}{2} \times (3.0 \times 10^{-2})^2]}$	
	$= 2.12 \times 10^{-2} \text{ m}$	A1
3(e)	flux linked to block changes / flux is cut by block which induces an e.m.f. in block	B1
	(eddy) currents induced in block cause heating	B1
	thermal / heat energy comes from (kinetic / potential) energy of oscillations / block	B1

34.

2(a)	e.g. period = 3 / 2.5	C1
	frequency = 0.83 Hz	A1
2(b)	light (damping)	B1
2(c)	at 2.7 s, $A_0 = 1.5$ (cm)	B1
	energy = $\frac{1}{2} m \times 4\pi^2 f^2 A_0^2$	B1
	$= \frac{1}{2} \times 0.18 \times 4\pi^2 \times 0.83^2 \times (1.5 \times 10^{-2})^2$	C1
	$= 5.51 \times 10^{-4}$ (J)	
	at 7.5 s, $A_0 = 0.75$ (cm)	B1
	energy = $\frac{1}{4} \times 5.51 \times 10^{-4}$ or energy = $\frac{1}{2} \times 0.18 \times 4\pi^2 \times 0.83^2 \times (0.75 \times 10^{-2})^2$	C1
	energy = 1.38×10^{-4} (J) change = $(5.51 \times 10^{-4} - 1.38 \times 10^{-4}) = 4.13$ J	A1

35.

3(a)(i)	e.g. period = 6 / 2.5	C1
	frequency = 0.42 Hz	A1
3(a)(ii)	energy = $\frac{1}{2} m \times 4\pi^2 f^2 y_0^2$	C1
	$= \frac{1}{2} \times 0.25 \times 4\pi^2 \times 0.42^2 \times (1.5 \times 10^{-2})^2$	C1
	$= 2.0 \times 10^{-4}$ J	A1
3(b)(i)	(induced) e.m.f. proportional to rate of change of magnetic flux (linkage) or cutting of magnetic flux	M1 A1
3(b)(ii)	coil cuts flux/field (of moving magnet) <u>inducing</u> e.m.f. in coil	B1
	(induced) current in resistor causes heating (effect)	M1
	thermal energy/heat derived from energy of oscillations (of magnet)	A1

36.

2(a)	(angular frequency =) $2\pi \times$ frequency or 2π /period	B1
2(b)(i)	1. displacement = 2.0 cm	A1
	2. amplitude = 1.5 cm	A1
2(b)(ii)	reference to displacement of oscillations or displacement from equilibrium position or displacement from 2.0 cm	B1
	straight line indicates acceleration $\propto$ displacement	B1
	negative gradient shows acceleration and displacement are in opposite directions	B1
2(b)(iii)	$\omega^2 = (-)1/\text{gradient}$ or $\omega^2 = (-)\Delta a/\Delta s$ or $a = (-)\omega^2 x$ <u>and</u> correct value of x	C1
	= e.g. (1.8 / 0.03) or (0.9 / 0.015) or (1.2 / 0.02) etc. or $0.9 = \omega^2 \times 0.015$	C1
	= 60	
	$f = \sqrt{60}/2\pi$ = 1.2 Hz	A1

37.

3(a)(i)	angle (subtended) where arc (length) is equal to radius	M1
	(angle subtended) at the centre of a circle	A1
3(a)(ii)	angular frequency = $2\pi \times$ frequency or 2π / period	B1
3(b)(i)	c/ML^3 is a constant so acceleration is proportional to displacement	B1
	minus sign shows that acceleration and displacement are in opposite <u>directions</u>	B1
3(b)(ii)	$c/ML^3 = (2\pi f)^2$	C1
	$c = 4\pi^2 \times 3.2^2 \times 0.24 \times 0.65^3$	C1
	$= 27 \text{ kg m}^3 \text{ s}^{-2}$	A1

38.

4(a)	frequency at which body will vibrate when there is no (resultant external) resistive force acting on it OR frequency at which body will vibrate when there is no driving force / external force acting on it	B1
4(b)(i)	resonance	B1
4(b)(ii)	peak is not sharp / peak not infinite height	M1
	so damped	A1
4(c)	e.g. (quartz crystal) to produce ultrasound (quartz crystal) in watch to keep timing NMR / MRI microwave ovens tuning circuits	B1

39.

2(a)(i)	straight line through origin indicates acceleration $\propto$ displacement	B1
	negative gradient shows acceleration and displacement are in opposite directions	B1
2(a)(ii)	$a = -\omega^2 y$ and $\omega = 2\pi f$	C1
	$4.5 = (2\pi \times f)^2 \times 8.0 \times 10^{-3}$ (or other valid read-off)	
	$f = 3.8 \text{ Hz}$	A1
2(b)(i)	maximum displacement upwards/above rest/above the equilibrium position	B1
2(b)(ii)	(just leaves plate when) acceleration = 9.81 ms^{-2}	C1
	$9.81 = (2\pi \times 3.8)^2 \times y_0$ or $9.81 = 563 \times y_0$	C1
	amplitude = 17 mm	A1

40.

4(a)	acceleration proportional to displacement	B1
	acceleration <u>directed</u> towards fixed point or displacement and acceleration in opposite <u>directions</u>	B1
4(b)(i)	1. amplitude decreases gradually so light damping or oscillations continue so light damping	B1
	2. loss of energy	B1
	due to friction in wheels or due to friction between wheels and surface (during slipping) or due to air resistance (on trolley)	B1
4(b)(ii)1.	$\omega^2 = 2k/m$	C1
	$= (2 \times 230) / 0.950$	C1
	$\omega = 22 \text{ rad s}^{-1}$	A1
4(b)(ii)2.	$T = 2\pi / \omega$	C1
	$T = (2\pi / 22) = 0.286 \text{ s}$	A1
	time $= 1.5 T$ $= 0.43 \text{ s}$	

41.

3(a)	$\omega^2 = 2g / L$	C1
	$T = 2\pi / \omega$	C1
	$\omega^2 = (2 \times 9.81) / 0.19$	A1
	$\omega = 10.2 \text{ (rad s}^{-1}\text{)}$	
	$T = 2\pi / 10.2$ $= 0.62 \text{ s}$	
3(b)(i)	e.g. viscosity of liquid/friction within the liquid/viscous drag/friction between walls of tube and liquid	B1
3(b)(ii)	(maximum) $\text{KE} = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$ or energy $= \frac{1}{2}m\omega^2 x_0^2$	C1
	change $= \frac{1}{2} \times 18 \times 10^{-3} \times 103 \times [(2.0 \times 10^{-2})^2 - (0.95 \times 10^{-2})^2]$	C1
	$= 2.9 \times 10^{-4} \text{ J}$	A1

42.

4(a)	(defining equation of s.h.m. is) $a = -kx$ where k is a constant or $a \propto -x$	B1
	g and L are constant (so $a \propto -x$ and hence s.h.m.)	B1
4(b)	$T = 0.50 \text{ s}$ and $T = 2\pi / \omega$	C1
	$\omega^2 = 2g / L$	C1
	$L = (2 \times 9.81 \times 0.50^2) / 4\pi^2$ $= 0.12 \text{ m}$	A1

4(c)(i)	Any one from: - viscosity of liquid - friction within the liquid - viscous drag - friction/resistance between walls of tube and liquid	B1
4(c)(ii)	(maximum) $KE = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$ or energy = $\frac{1}{2}m\omega^2 x_0^2$	C1
	ratio = $(1.3/2.0)^2$ = 0.42	A1

43.

3(a)(i)	mention of upthrust and weight	B1
3(a)(ii)	upthrust is greater than the weight	B1
	(resultant force is) upwards	B1
3(b)	A, ρ, g and M are constant	B1
	either acceleration $\propto$ – displacement	B1
	or acceleration $\propto$ displacement and (– sign indicates) a and x in opposite directions	
3(c)(i)	either $\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$\omega = 2\pi / 1.3$ = 4.8 rad s ⁻¹	A1
3(c)(ii)	$\omega^2 = A\rho g / m$	C1
	$4.83^2 = (4.5 \times 10^{-4} \times \rho \times 9.81) / 0.17$	C1
	$\rho = 900 \text{ kg m}^{-3}$	A1

44.

3(a)(i)	amplitude = 0.020 m	A1
3(a)(ii)	$T = 0.60 \text{ s}$	C1
	$f = 1 / T$ = 1.7 Hz	A1
3(a)(iii)	$a = (-)\omega^2 x$ and [$\omega = 2\pi f$ or $\omega = 2\pi / T$]	C1
	$a = (4\pi^2 / 0.60^2) \times 2.0 \times 10^{-2}$ = 2.2 m s ⁻²	A1
3(b)	$1.67 = (1 / 2\pi) \times [(24 \times 10^{-4} \times \rho \times 9.81) / 0.23]^{1/2}$	C1
	$\rho = 1.1 \times 10^3 \text{ kg m}^{-3}$	A1
3(c)	wave starting with a peak at (0,6)	B1
	wave with same period (or slightly greater)	B1
	peak height decreasing successively	B1

45.

3(a)(i)	0.10 s or 0.30 s or 0.50 s or 0.70 s or 0.90 s	A1
3(a)(ii)	0 or 0.40 s or 0.80 s	A1
3(b)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / 0.40$	A1
	$= 16 \text{ rad s}^{-1}$	
3(b)(ii)	$v_0 = \omega x_0$	C1
	$= 15.7 \times 2.5 \times 10^{-2}$	A1
	$= 0.39 \text{ m s}^{-1}$	
	or	
	tangent drawn at steepest part and working to show attempted calculation of gradient	(C1)
	leading to $v_0 = 0.39 \text{ m s}^{-1}$ (allow $\pm 0.15 \text{ m s}^{-1}$)	(A1)
3(b)(iii)	$a_0 = \omega^2 x_0$	C1
	$a_0 = (15.7^2 \times 2.5 \times 10^{-2})$	A1
	$= 6.2 \text{ m s}^{-2}$	
	or	
	$a_0 = \omega v_0$	(C1)
	$a_0 = 15.7 \times 0.39$	(A1)
	$= 6.2 \text{ m s}^{-2}$	
3(c)	period is shorter/lower	B1
	Any one from: - greater spring constant/stiffness (restoring) force is greater (for any given extension) - acceleration is greater (for any given extension) - greater energy/maximum speed (for a given amplitude)	B1

46.

4(a)(i)	distance from a (reference) point in a given direction	B1
4(a)(ii)	line is not straight or gradient is not constant	B1
4(b)(i)	0.85–0.90 cm	A1
4(b)(ii)	$a = -(2\pi f)^2 x$	C1
	e.g. $1.2 = 4\pi^2 \times f^2 \times (0.90 \times 10^{-2})$	C1
	$f = 1.8 \text{ Hz}$	A1
4(c)	complete circle/ellipse enclosing the origin	B1
	closed shape passing through $(0, \pm v_0)$ and $(\pm x_0, 0)$	B1

47.

4(a)(i)	loss of energy	B1
4(a)(ii)	amplitude (of oscillations) decreases (with time)	B1
4(b)(i)	$\omega = 2\pi / T$	C1
	$T = 0.80 \text{ s}$, so $\omega = 2\pi / 0.80$	A1
	$\omega = 7.9 \text{ rad s}^{-1}$	
4(b)(ii)	$\omega^2 = 2k / M$	C1
	$7.9^2 = 2k / 1.2$	A1
	$k = 37 \text{ N m}^{-1}$	
4(c)(i)	(one) smooth curve, not touching the f -axis, with two concave sides meeting at a peak in between them	B1
	(one) peak at 1.0ω	B1
4(c)(ii)	- lower peak/(whole) line is lower - flatter <u>peak/peak</u> is less sharp - peak at (slightly) lower angular frequency/peak moves to left any two points, one mark each	B2

48.

3(a)(i)	0.050 m	A1
3(a)(ii)	$\omega = v_o / x_o$	C1
	$T = 2\pi / \omega$	C1
	$0.42 = (2\pi \times 0.050) / T$	
	$T = 0.75 \text{ s}$	A1
3(a)(iii)	one point labelled P where ellipse crosses displacement axis marked	A1
3(b)(i)	(induced) e.m.f. proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
3(b)(ii)	(there is) current in the circuit	B1
	either	
	current causes thermal energy (dissipated) in resistor	B1
	thermal energy comes from energy of magnet	B1
	or	
	current causes magnetic field around coil	(B1)
	two fields cause an opposing force on magnet	(B1)

49.

3(a)(i)	amplitude = 4.9 cm	A1
3(a)(ii)	frequency = $2700 / 60$ = 45 Hz	A1
3(a)(iii)	$v_0 = x_0 \omega$ and $\omega = 2\pi f$	C1
	$v_0 = 4.9 \times 10^{-2} \times 2\pi \times 45$ = 14 m s^{-1}	A1
3(a)(iv)	$v = \omega (x_0^2 - x^2)^{1/2}$ = $2\pi \times 45 \times [(4.9 \times 10^{-2})^2 - (2.6 \times 10^{-2})^2]^{1/2}$ = 12 m s^{-1}	C1
		A1
3(b)	$F = ma$ and $a_0 = v_0 \omega$ or $a_0 = x_0 \omega^2$	C1
	$F = 0.64 \times 13.9 \times 2\pi \times 45$ or $0.64 \times 4.9 \times (2\pi \times 45)^2$	C1
	= 2500 N	A1

50.

4(a)	$(\omega = 2\pi / T \text{ and } T = 2.2 \text{ s so})$ $\omega = 2\pi / 2.2 = 2.9 \text{ rad s}^{-1}$	A1
4(b)(i)	$\omega^2 = g / R$	C1
	$R = 9.81 / 2.86^2$ = 1.2 m	A1
4(b)(ii)	$v_0 = \omega x_0$	C1
	= $2.9 \times 3.0 \times 10^{-2}$ = 0.087 m s^{-1}	A1
4(c)	smooth wave starting at 3.0 cm when $t = 0$	B1
	positions of peaks and troughs show same period (or slightly longer)	B1
	each peak and trough at lower amplitude than the previous one	B1

51.

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration in opposite <u>direction</u> to displacement or acceleration (directed) towards equilibrium position	B1
3(b)	$v = \omega (x_0^2 - x^2)^{1/2}$ and $\omega = 2\pi f$ or $v_0 = x_0 \omega$ and $\omega = 2\pi f$	C1
	substitution of any correct point from graph, e.g. for $x = 0$: $0.25 = 2\pi f \times 8.8 \times 10^{-2}$ $f = 0.45 \text{ Hz}$	C1
		A1
3(c)	$1 / 0.45 = 2\pi \times (L / 9.81)^{1/2}$	C1
	$L = 1.2 \text{ m}$	A1
3(d)	ellipse about the origin with same intercepts on x-axis	B1
	ellipse about the origin crossing v-axis inside original loop	B1

52.

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration is in opposite <u>direction</u> to displacement or acceleration is (directed) towards a fixed point	B1
3(b)(i)	zero	B1
3(b)(ii)	E_T is maximum potential energy = mgh $E_T = 94 \times 10^{-3} \times 9.81 \times 0.90 \times 10^{-2}$ $= 8.3 \times 10^{-3} \text{ J}$	C1 A1
	$E_{\text{MAX}} = \frac{1}{2} m v_0^2$ and $v_0 = \omega x_0$ or $E_{\text{MAX}} = \frac{1}{2} m (\omega x_0)^2$ $8.3 \times 10^{-3} = \frac{1}{2} \times 94 \times 10^{-3} \times \omega^2 \times (12.7 \times 10^{-2})^2$...leading to $\omega = 3.3 \text{ rad s}^{-1}$	C1 A1
3(c)	$T = 2\pi / \omega$	C1
	$2\pi / 3.3 = 2\pi \times (L / 9.81)^{1/2}$	C1
	$L = 0.90 \text{ m}$	A1

53.

4(a)	acceleration and displacement are in opposite directions	B1
4(b)(i)	$F = kx$ $= 8.0 \times (0.060 - 0.048)$ or $8.0 \times (0.060 + 0.048)$ or 8.0×0.012 or 8.0×0.108	M1
	$\Sigma F = (8.0 \times 0.012) - (8.0 \times 0.108) = 0.77 \text{ N}$ or $\Sigma F = 0.864 - 0.096 = 0.77 \text{ N}$	A1
4(b)(ii)	$a = \frac{F}{m}$ $= \frac{0.77}{0.25}$ $= 3.1 \text{ m s}^{-2}$	A1
4(b)(iii)	$a = -\omega^2 x$ $\omega = \sqrt{\frac{3.1}{0.048}}$ $\omega = 8.04$	C1
	$T = 2\pi / \omega$	C1
	$T = 2\pi / 8.04$ $= 0.78 \text{ s}$	A1
4(b)(iv)	(resultant) force halved and distance halved	B1
	same T	B1

54.

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration is in opposite <u>direction</u> to displacement	B1
3(b)	$\omega^2 = 2k / m$ and $\omega = 2\pi f$	C1
	$(2\pi f)^2 = (2 \times 130) / 0.84$	C1
	$f = 2.8 \text{ Hz}$	A1
3(c)(i)	resonance	B1
3(c)(ii)	oscillator supplies energy (continuously)	B1
	energy of trolley constant so energy must be dissipated or without loss of energy the amplitude would continuously increase	B1

55.

3(a)	acceleration in opposite <u>direction</u> to displacement shown by – sign	B1
	g / L is constant	M1
	(so) acceleration is (directly) proportional to displacement	A1
3(b)	$\omega^2 = g / L$	C1
	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$(2\pi / T)^2 = 9.81 / 0.18$	A1
	$T = 0.85 \text{ s}$	
3(c)	energy $\propto x_0^2$	C1
	(after 3 cycles,) amplitude = $(0.94)^3 x_0$ = $0.83 x_0$	C1
	ratio final energy / initial energy = 0.83^2 = 0.69	A1

56.

4(a)(i)	5.0 cm	A1
4(a)(ii)	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$\omega = 2\pi / 4.0$ = 1.6 rad s^{-1}	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	= 1.57×5.0	A1
	= 7.9 cm s^{-1}	
4(b)	- initial pull was to the right - distance from X to trolley (at equilibrium) is 20 cm - period is 4.0 s - initial motion undamped - motion becomes damped at/from 12 s - damping is light - maximum speed at 1 s, 3 s, etc. / stationary at 2 s, 4 s, etc. Any three points, 1 mark each	B3

4(c)	sketch: closed loop encircling (20, 0)	B1
	minimum L shown as 15 cm <u>and</u> maximum L shown as 25 cm	B1
	minimum v shown as -7.9 cm s^{-1} <u>and</u> maximum v shown as $+7.9 \text{ cm s}^{-1}$	B1

57.

4(a)	straight line through the origin	B1
	negative gradient	B1
4(b)	$a = (-)\omega^2 x$ <u>and</u> $T = 2\pi / \omega$	C1
	e.g. $\omega = \sqrt{(0.80 / 0.12)}$ (any correct pair of values of a and x) ($= 2.58 \text{ rad s}^{-1}$)	C1
	$T = 2\pi / 2.58$ $= 2.4 \text{ s}$	A1
4(c)(i)	Point labelled P at one end of the line	B1
4(c)(ii)	Point labelled Q at displacement with magnitude more than half but less than maximum	B1

58.

3(a)	upthrust, weight	B1
3(b)	upthrust greater than weight so (resultant force is) upwards	B1
3(c)(i)	A , g and ρ all constant so $F \propto x$	B1
	minus sign means F and x are in opposite directions	B1
3(c)(ii)	$(a = \frac{F}{m} \text{ so } a = (-)\frac{A\rho g x}{m})$	M1
	so $\omega^2 = \frac{A\rho g}{m}$ hence $\omega = \sqrt{\frac{A\rho g}{m}}$	A1
3(d)(i)	damping due to viscous forces	B1
3(d)(ii)	$(E =) \frac{1}{2} m \omega^2 x_0^2$	C1
	$\omega^2 = (-)$ gradient	C1
	$(E =) \frac{1}{2} m \omega^2 (x_1^2 - x_2^2)$ $= \frac{1}{2} \times 0.57 \times (2.3 / 0.020)(0.020^2 - 0.016^2)$ $= 4.7 \times 10^{-3} \text{ J}$	A1

59.

4(a)	straight line through origin shows that a is proportional to x	B1
	negative gradient shows that a is in opposite direction to x	B1
4(b)(i)	$a_0 = \omega^2 x_0$ or $a = -\omega^2 x$ or $\omega^2 = -\text{gradient}$	C1
	$\omega = \sqrt{(0.40 / 0.050)}$ $= 2.8 \text{ rad s}^{-1}$	A1
4(b)(ii)	$k = \omega^2 L$ $= 2.8^2 \times 1.24$	C1
	$= 9.7 \text{ m s}^{-2}$	A1
4(c)	(increasing L causes) ω to decrease or energy $(= \frac{1}{2} m \omega^2 x_0^2) = \frac{1}{2} m k x_0^2 / L$ (and L increases)	M1
	so amplitude increases	A1

60.

4(a)	oscillations (of object) at maximum amplitude	B1
	when driving frequency equals natural frequency (of object)	B1
4(b)(i)	$T = 2\pi / \omega$	C1
	$= 2\pi / 5.0\pi$ $= 0.40 \text{ s}$	A1
4(b)(ii)	displacement scale labelled $-1.0, -0.5, (0), 0.5, 1.0$ on the 2 cm tick marks	B1
	t scale labelled $0.2, 0.4, 0.6, 0.8, 1.0, 1.2$ on the 2 cm tick marks	B1
4(b)(iii)	$\phi = 2\pi \Delta t / T$ $= 2\pi \times 0.10 / 0.40$ or $2\pi \times 0.30 / 0.40$	C1
	$= 1.6 \text{ rad}$ or 4.7 rad	A1

61.

3(a)	$a = -\omega^2 x$	M1
	a = acceleration, x = displacement from equilibrium position and ω = angular frequency	A1
3(b)(i)	$x_0 = 0.12 \text{ m}$	A1
3(b)(ii)	$v = \omega \sqrt{(x_0^2 - x^2)}$ two (x, v) pairs correctly read from Fig. 3.2 (one may be $(x_0, 0)$ or value of x_0 from (i))	C1
	e.g. $0.20 = \omega \sqrt{(0.12^2 - 0)}$ leading to $\omega = 1.7 \text{ rad s}^{-1}$	A1

3(b)(iii)	$E = \frac{1}{2}M\omega^2 x_0^2$	C1
	$0.050 = \frac{1}{2} \times M \times 1.67^2 \times 0.12^2$	A1
	$M = 2.5 \text{ kg}$	
	or	
	$(E_K)_{\max} = \frac{1}{2}Mv_0^2$	(C1)
	$0.050 = \frac{1}{2} M \times 0.20^2$	(A1)
	$M = 2.5 \text{ kg}$	
3(c)(i)	loss of (total) energy (of system)	B1
	due to resistive forces	B1
3(c)(ii)	closed loop surrounding the origin with maximum x at $\pm 0.060 \text{ m}$ passing through $v = 0$	B1
	maximum velocity shown as $\pm 0.10 \text{ m s}^{-1}$ passing through $x = 0$	B1

62.

4(a)(i)	$x_0 = 8.0 \text{ cm}$	A1
4(a)(ii)	$\omega = 2\pi / T$	C1
	$= 2\pi / 4.0 = 1.6 \text{ rad s}^{-1}$	A1
4(a)(iii)	$E = \frac{1}{2}m\omega^2 x_0^2$	C1
	$= \frac{1}{2} \times 36 \times 1.6^2 \times 0.080^2$	C1
	$= 0.29 \text{ J}$	A1
4(b)	dome-shaped curve, starting and ending at $E_K = 0$	B1
	maximum E_K shown as 0.29 J	B1
	position of peak shown at $h = 10.0 \text{ cm}$	B1
	line intercepts h -axis at $h = 2.0 \text{ cm}$ and at $h = 18.0 \text{ cm}$	B1

63.

3(a)	P: total energy Q: potential energy R: kinetic energy	B2
3(b)	$E = \frac{1}{2}m\omega^2 x_0^2$ or $E = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$	C1
	$6.4 \times 10^{-3} = \frac{1}{2} \times 0.130 \times \omega^2 \times 0.015^2$ ($\omega^2 = 438$) ($\omega = 20.9$)	C1
	$T = 2\pi / \omega$	C1
	$= 2\pi / 20.9$	A1
	$= 0.30 \text{ s}$	
3(c)(i)	resistive forces	B1
3(c)(ii)	0.92^5	C1
	decrease in energy $= 6.4 - (6.4 \times 0.92^5)$ $= 2.5 \text{ mJ}$	A1
3(c)(iii)	light damping because the amplitude of oscillations gradually reduces or light damping because the system still oscillates	B1

64.

4(a)(i)	$\omega = 2\pi f$	C1
	$f = 9.7 / 2\pi$	A1
	$= 1.5 \text{ Hz}$	
4(a)(ii)	amplitude $= \sqrt[3]{11.6} = 3.4 \text{ cm}$	A1
4(a)(iii)	$a_0 = \omega^2 x_0$	C1
	$= 9.7^2 \times 3.4 \times 10^{-2}$	A1
	$= 3.2 \text{ m s}^{-2}$	
4(b)	sketch: straight line through the origin with negative gradient	B1
	line with negative gradient passing through $(+3.4, -a_0)$ and $(-3.4, +a_0)$	B1
	line with ends at $x = \pm 3.4 \text{ cm}$ and $a = \pm a_0$	B1
4(c)	sum of potential energy and kinetic energy is constant	B1
	at maximum displacement, kinetic energy is zero or at maximum displacement, potential energy is maximum	B1
	at zero displacement, kinetic energy is maximum or at zero displacement, potential energy is minimum	B1

65.

4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) directed towards a fixed point	B1
4(b)(i)	$\omega = 2\pi / T$	C1
	$\omega = 2\pi / 3.0$	A1
	$= 2.1 \text{ rad s}^{-1}$	
4(b)(ii)	$E = \frac{1}{2} m \omega^2 x_0^2$	C1
	$= \frac{1}{2} \times 0.81 \times 2.1^2 \times 0.036^2$	A1
	$= 2.3 \times 10^{-3} \text{ J}$	
4(c)	sketch: line starting at $(0, 0.036)$ and not reaching $x = \pm 0.036 \text{ m}$ at any other time	B1
	smooth curve, with no sudden changes in gradient, showing continuously decreasing magnitude of x from maximum displacement at $t = 0$ to final displacement of zero where the gradient is also zero	B1
	displacement reaches final value of zero between $t = 0.75 \text{ s}$ and $t = 3.0 \text{ s}$ at the latest	B1

66.

4(a)(i)	amplitude = $\frac{1}{2} \times 7.2 \times 10^{-15}$ = $3.6 \times 10^{-15} \text{ m}$	A1
4(a)(ii)	$\omega = 2\pi / (0.20 \times 10^{-6})$ = $3.1 \times 10^7 \text{ rad s}^{-1}$	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	$v_0 = 3.1 \times 10^7 \times 3.6 \times 10^{-15} = 1.1 \times 10^{-7} \text{ m s}^{-1}$	A1
4(b)(i)	$I_0 = nA v_0 e$ = $8.5 \times 10^{28} \times 4.3 \times 10^{-4} \times 1.1 \times 10^{-7} \times 1.60 \times 10^{-19}$ = 0.64 A	C1
		A1
4(b)(ii)	sketch: two cycles of sinusoidal curve of amplitude I_0 and period $0.20 \mu\text{s}$	B1
	correct phase, with $I = +I_0$ at $t = 0$	B1
4(b)(iii)	equation of form $I = I_0 \cos \omega t$	M1
	value of I_0 used matches answer to (b)(i) and value of ω used matches answer to (a)(ii) [if (a)(ii) and (b)(i) correct then $I = 0.64 \cos (3.1 \times 10^7 t)$]	A1
4(b)(iv)	$I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ = $0.64 / \sqrt{2}$ = 0.45 A	A1